

Title: Metal/Semiconductor phasetransitions in TMDCs

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Summary

The properties of atomically thin materials depend dramatically on their lattice structures. Alterations in the atomic positions in such materials, that are nothing much than a surface, lead to a drastic change in their physical properties. It is somewhat fundamental to understand what is the preferred structure of a two-dimensional material. More interesting it is the fact that some two-dimensional materials exist in different phases, they share the same stoichiometry, but they have different properties. Recently has been shown that there is the possibility to follow the phasetransition from a metallic phase to a semiconductive phase in MoS₂, one of the most promising two-dimensional material. This opens a new way to control the physical properties at the microscopic scale and start the new field of phase engineering for two-dimensional materials. Here we propose to study systematically these phasetransitions in TMDCs, switching the focus more than on the transition itself compare to the particular material. At the practical level, we propose to study in detail the known phasetransitions and try to apply different methods to realize the transition. After that and with knowledge gained we propose to investigate not yet observed phase transitions, by following the predictions from ab initio highthroughput simulations, with the ultimate goal of understanding whether phase engineering can be used to control the physical properties of two-dimensional materials.

Introduction

The discovery of the possibility to exfoliate graphite and producing graphene flakes¹ officially started the research for two-dimensional material. After that many two-dimensional materials were discovered and they were formalized them in groups². The family of transition-metal dichalcogenides started to attract a lot of attention after the discovery of MoS₂ in 2010³. In the following years, other materials were added to this group⁴, just by moving the transition metal and chalcogen on the rows and columns of the periodic table. At the same time the unique properties and the possible applications of TMDCs were growing at the same rate. Unlike graphene, the semiconductive nature of MoS₂, opens up more applications and challenges in electronics⁵, optoelectronics⁶ and catalysts⁷. This naturally led to use them in the challenge to develop an atomically thin transistor⁵.

This idea, which is the main driving force in two-dimensional material research, rely on the ability to fast switch the conductive state of the material. Two-dimensional material, due to their microscopic nature, seem suitable for this scope. Recently there are some promisingly observations to achieve this through the phase engineering of the material⁸.

It is known that MoS₂, the prototypical TMDCs, has an hexagonal structure⁹, usually called the 2H phase. This crystal structure gives the unique property to MoS₂ that makes suitable for optoelectronic applications because of the finite bandgap. At the same time MoS₂ exhibits a metastable trigonal phase called the 1T phase shown in Figure (1).

Despite the 2H phase is the most stable one, it is possible to make the 1T stable for example with: Li-ion intercalation¹⁰, strain¹¹, electron beam¹² and argon bombardment¹³. More interestingly there is also the possibility to follow in real time the transition between the two phases⁸.

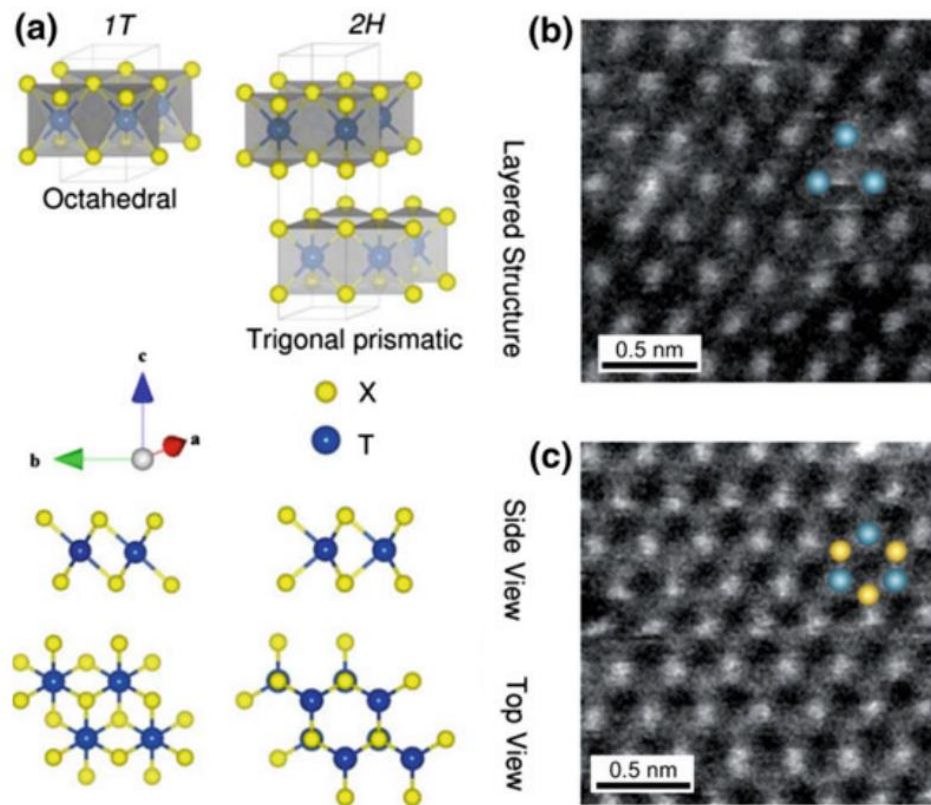


Figure 1. The 2H and 1T phase for a general TMDCs. On the right, atomic resolution STEM images of 1T (b) and 2H (c) for monolayer MoS₂.¹⁴

Going beyond the case of MoS₂, in the family of TMDCs some materials follow the case discussed before. Indeed just by slightly moving on the periodic table, or for the transition metal or for the chalcogen, the H phase is always the most stable phase. This is the case for the sulfide and selenide or when the transition metal used is Tungsten, while for telluride the most stable phase is the 1T phase. Exploring all the possible combinations of the experimentally known TMDCs there is always the competition between these two phases¹⁴.

Hypothesis & goals

As largely discussed in the previous section, the metal/semiconductor phase transition in MoS₂, it is an interesting method to control the transport properties and it opens a lot of questions. In particular the main question that the project wants to answer is whether it is possible to use phase engineering of TMDCs to control the conductive state of the material.

It would be fascinating being able to control the properties of the device from outside and it is also what research is trying to accomplish with two-dimensional material. This will be done first by trying to study in details existing well known cases in which the transition happens and investigate if there is something not yet clear in the mechanism behind the transition.

Whether it is possible to use the transition in applications in MoS₂, the idea is still promising because of the large number of TMDCs. Indeed it is possible to explore different combination of these materials in order to find which material is more suitable for phase engineering applications. The exploration of the different combination will be supported by the predictions of first principles simulations. The project will answer also if it is possible to use ab initio highthroughput simulations to guide the discovery of new phase transitions in two-dimensional materials.

Relevance/Impact

At the end of the work, the project will give a more clear picture of the status of phase transitions in TMDCs. Indeed so far, much attention was given to the particular material and its properties in order to find applications. Here we want to switch the focus from the material to the transition itself as tool to tailor the physical properties of the materials. After the end of the project will emerge if phase engineering of TMDCs is a feasible way to control the conductive states of the material.

At the same time the extensive investigations performed during the project will give information if there are any other TMDCs in which it is possible to stabilize a metastable phase, like in the case of the 1T phase in MoS₂.

In addition trying to study the phase transitions in several materials will give more insight behind the mechanism that drives the transformation.

Scientific background

A way to understand the mechanism behind the phase transition is usually given by first principles simulations. Indeed in recent years a lot of effort was put to calculate the properties, but also to predict new two-dimensional materials. There are indeed different results, using ab initio methods, that are trying to find a definition for the stability of two-dimensional materials that match the experimental observation¹⁵.

Two ways are commonly accepted ways to define the stability. One that comes from more thermodynamic consideration by comparing the total energy obtained from the ab initio calculation with the total energies of isolated phases of the elements of the material. This quantity is usually called the heat of formation, ΔH , therefore defined as the energy of the material with respect to the standard states of its constituent elements. For example, the heat of formation per atom of a binary compound, A_xB_y is $(E(A_xB_y) - xE(A_x) - yE(B_y)) / (x + y)$.

The other way is to consider if the material is unstable against some distortion and this brings to a change of the phase. This is defined with dynamical stability, which basically identifies how likely the material wants to make a distortion. In the calculations this is obtained by looking in negative frequencies in the phonons bands structure.

It is shown that these definitions give results in agreement with the observed experimental materials. There are of course some corner cases due both to the theory behind the ab initio method, but also with approximation schemes used to evaluate the calculation in practice.

For example we report the two phases of MoS₂, that it is largely discussed in the previous sections. One of the well accepted ways to assess the instability in this case is by looking at the phonons bands. In the figure (2) the difference between the two phases are shown, with large negative frequencies at the point M and K of the Brillouin zone.

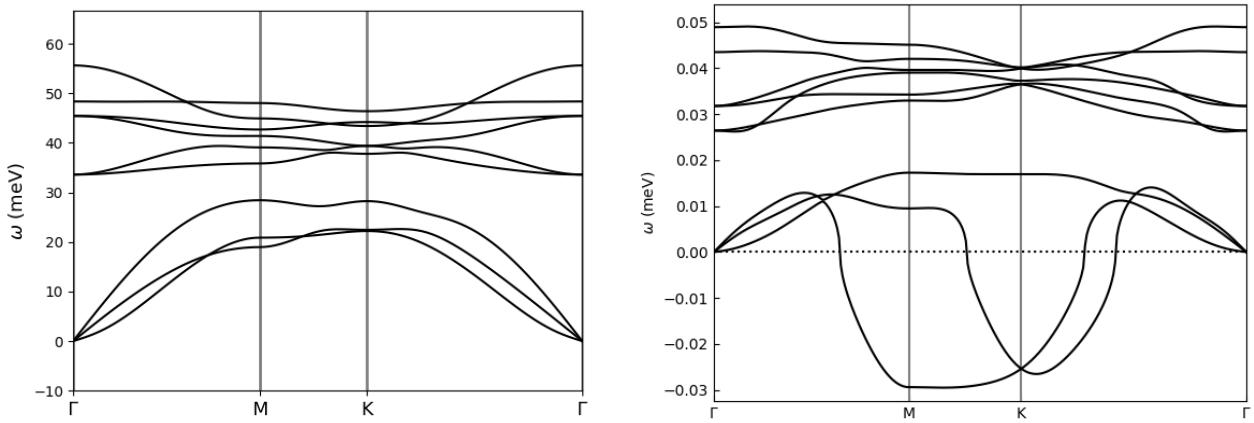


Figure 2. Differences in the phonons bands structure between the 2H-phase (left) and 1T-phase (right).

Methodology and plan

As largely discussed before, the aim of the project is to investigate if the phase transitions in TMDCs can be used for applications. In order to accomplish this, the first part of the project is to start from the well known example of MoS₂. This means in particular to reproduce the transition and follow it in real time.

In the second part of the project, some of the most promising materials are studied, following the prediction from *ab initio* simulation.

Both stages can be divided in two parts. The first is the fabrication and synthesis of the material, which will follow the standard approach used for two-dimensional materials¹⁶. In the second part the properties of the two phases will be studied. This will be done with the ARPES for electronic properties, the STM for structural characterization of the two phases. This will be used also to try to follow the transition in real time as done in a previous work⁸.

Finally if some promising transition is found this will be studied by looking at the transport properties.

Self-analysis/feasibility

The answer that project wants to find is at a quite fundamental level. As mentioned before, the project wants to address if phase engineering is way to control the properties of the material.

The stability of the two phases connected by the phase transition will play a big role in the project. For example one of the problem could be that the phase of the material is not stable in the presence of Oxygen. This problem can be solved with encapsulation in BN. It is also shown that this do not prevent from studying the structure of the material with STM¹⁷. Therefore even if the material is inside one or few layer of BN, it is still possible to characterize their structural properties.

Phase	Milestone	Risk/contingency
Phase 1 M1-12	M1.1. Start working with MoS2 and reproduce the phase transition from the 2H to the 1T ⁸ .	R1.1. It is hard to follow the transition in real time.
Phase 2 M13-24	M2.1 Studying the transition from most promising candidates	R2.1 Some of the material are not stable because they oxidize in the presence of oxygen → Contingency: try the encapsulation with BN to protect the material from oxidization.
Phase 3 M25-36	M3.1 Focus on the most promising materials .	R3.1 No material has a stable phase → Contingency: Go back to the existing cases

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